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PI Name:	Fornace, Albert M.D.		
Project Title:	Mouse models approach for intestinal tumorigenesis estimates by space radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
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Space Biology Special Category:	None		
PI Email:	af294@georgetown.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	202 687-7843
Organization Name:	Georgetown University		
PI Address 1:	Dept. of Oncology, Lombardi Comprehensive Cancer Center		
PI Address 2:	Research Building, Room E504, 3970 Reservoir Rd., NW		
PI Web Page:			
City:	Washington	**State:**	DC
Zip Code:	20007-2126	**Congressional District:**	1
Comments:	http://www9.georgetown.edu/		
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No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
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Key Personnel Changes/Previous PI:			
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Task Description:	Ionizing radiation (IR) is a known risk factor for colorectal (CR) cancer. Considering the high spontaneous incidence of gastrointestinal (GI) cancer in the US and even higher incidence of pre-malignant lesions, such as colonic polyps, an even modest increase by radiation exposure could have a significant effect on health risk estimates for manned space flight. During space travel, astronauts are exposed to levels of radiation, which increases their risk for acute short-term deleterious effects, such as during a solar storm, and for long-term deleterious effects like cancer. Long duration space missions, like those at the International Space Station, lunar missions, or a mission to Mars, are of sufficient duration to allow for appreciable cumulative doses of cosmic radiation that could have profound health consequences. This project is important because there are no good estimates for intestinal, particularly CR, cancer by space radiation. Considering the increased CR cancer incidence in A-bomb survivors and the novel characteristics of space radiation, cancer causation by space radiation could potentially be even greater. The overall goal of this project is to improve NASA risk estimates for space radiation-induced intestinal tumors and to investigate modulation of specific signaling pathways as a model for cancer prevention. We are employing APC mouse models that have been successfully used to demonstrate radiation-induced intestinal cancer. The APC models have inactivation of one APC (adenomatous polyposis coli) gene and the mice develop intestinal tumors, adenomas. Radiation can inactivate the remaining APC gene and thus initiate increased intestinal tumor development. The strategy is to compare results for space radiation, protons and HZE (56Fe), on intestinal tumor development to that of low-LET (linear energy transfer) radiation (gamma rays). Since low-LET risk estimates are available for human populations, we will then be able to model the relative risk for space radiation. In particular, this ratio can then be extrapolated to human exposure to low-LET radiation to then make an estimate of risk for humans exposed to space radiation. Many of the molecular events leading to intestinal tumors have been characterized. We will monitor and compare molecular changes in three types of radiation-induced tumors. The specific aims for this proposal are: 1. Quantitatively compare HZE ion or high-energy proton irradiation to gamma-ray effects on intestinal tumor incidence and grade using the Apc+/min tumor model. 2. Characterize and compare inactivation of wt APC locus on chromosome 18B1 after gamma vs. HZE ion or high-energy proton irradiation with the Apc+/min tumor model. 3. Determine intestinal tumor incidence in the Apc+/min; Ppm1d-/- mouse model and mechanisms for reactivation of the Apc+/min tumor phenotype by radiation. 4. Determine molecular signatures by comparative genomic hybridization (CGH) and expression profiling in spontaneous intestinal tumors compared to those induced by HZE ions, high-energy protons, and gamma irradiation. 5. Quantitatively compare HZE ion or high-energy proton irradiation vs. gamma-ray effects on incidence of other tumors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gastrointestinal tumors are frequent in the U.S, and the incidence of CR cancer in 2005 (latest year for which statistics are available) was 56.4 per 105 with a death rate of 21.0 per 105 or about 53,005 deaths per year. Colorectal cancer when considered in men and women is the second leading cancer killer in the United States (http://www.cdc.gov/). Considering the high frequency of CR tumors in the American population, an even small increase by space radiation could have a major impact on risk estimates and planning of future manned space missions. The effect of complex DNA lesions, which are produced by HZE radiation, on intestinal tumorigenesis may provide insight into mutagenic processes affecting genome integrity and CR carcinogenesis.
Task Progress:	The overall plan is to apply the well-characterized APC mouse models (APC+/min and APC1638N) of intestinal tumorigenesis for development of risks estimates for various radiation exposures during manned space flight. To undertake a comparative study of intestinal tumor risk in mice following exposure to different types of radiation, the relative toxicity of 56Fe particles and protons to gamma-rays needed to be determined so that we can standardize with doses equitoxic to 2 and 5 Gy of gamma-radiation in C57Bl/6J mice; these gamma-ray doses have been shown to be effective in increasing intestinal tumorigenesis. The rationale for this initial study is that acute toxicity for the planned radiation sources in mice is not available. We have carried out four runs so far at the NSRL at Brookhaven National Laboratory. Results to date indicate that the relative biological effectiveness (RBE) for 56Fe particles compared to gamma-rays is 1.25 and RBE for proton is 1.06. Although the RBE for protons was expected, the RBE for 56Fe particles is surprisingly low and is an important parameter that will allow us to plan long-term experiments. Irradiated mice are being observed long-term, and specific tissues are also being provided to various NASA investigators to complement their own projects. Subsequently we irradiated 6 to 8 week old female APC+/min mice with iron particles (4 Gy of 1 GeV/n 56Fe) and protons (4.76 Gy of 1 GeV/n) which is equivalent to 5 Gy of gamma-radiation. Including gamma-rays a total of 100 mice were irradiated and have been monitored for intestinal tumorigenesis. Mice were observed for 100 days and at the end of the observation period mice are being humanely euthanized as per Georgetown University Animal Care and Use Committee (GUACUC) approved protocol and tumors counted and tumor samples harvested for qualitative analysis. The APC+/min mice show frequent spontaneous tumors and IR typically triggers a modest GI tumor increase in adults compared to larger increases in newborn or younger mice, and we also found a modest but significant increase after gamma-rays. In contrast, the increase in tumor number with equitoxic doses of protons or Fe ions radiation was significantly greater. Interestingly, in the course of our analyses, we also noted a larger increase of clustered tumors defined here as five or more adenomas without intervening normal mucosa with space radiation compared to gamma rays. Results to date indicate the space radiation, both high-energy protons and 56Fe particles, may have unique properties conferring an increased tumorigenic risk compared to gamma rays. Quantitative study with 5 Gy gamma and equitoxic doses of proton and iron has been completed and work on 2 Gy gamma and equitoxic doses of proton and iron is ongoing. Tumor samples harvested during quantitative study are now being processed for qualitative analysis. Quantitative and qualitative study on intestinal tumorigenesis is also performed in APC1638N/+ with the same radiation dose as mentioned earlier. Six- to eight-week old APC1638N/+ male and female mice were exposed whole-body to 5 Gy of gamma-rays, 4.76 Gy of 1 GeV/n protons or 4 Gy of 1 GeV/n 56Fe. Age-matched controls were left untreated. Mice were observed for 150 days post-radiation and at the end of the observation period mice are being humanely euthanized as per Georgetown University Animal Care and Use Committee (GUACUC) approved protocol and tumors counted and tumor samples harvested for qualitative analysis. Age matched controls were included for all strains and

	radiation treatments. Quantitative study with gamma radiation is complete and we observed a significant increase in total intestinal tumor burden following exposure to 5 Gy of gamma-rays. Importantly, unlike APC min/+, large intestinal tumor multiplicity were greater in APC1638N/+ and we report here that a single adenoma was found in the large intestine of only one age-matched control, while gamma-irradiated Apc1638N mice showed an average of 4.7 ± 2.0 adenomas (range 2-8 tumors) in the colon and, therefore, a substantial 47-fold increase compared to unirradiated animals. Studies with space radiation in these mice are ongoing and preliminary results indicate a significant increase in intestinal tumor formation. Considering the large radiation-induced relative increase in tumors in the APC1638N/+, low dose (<1 Gy) studies are now feasible and are a priority in future studies.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
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